

Vacua search via metaheuristics in k-theory model

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Abstract

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1 Introduction

In quantum physics there are several approaches to study the behaviour of subatomic elements in an attempt to understand the fundamental interactions in the universe.

In quantum field theory, there are some problems that come up when trying to analyze dome phenomena at subatomic interaction level. String theory is an approach intended to build mathematically feasible models of quantum *gravity*[1] by understanding fundamental particles as strings instead of puntual elements.

The interest in string theory is that from it emerges the relativist behaviour leading to the possibility of building the *so-called theory of everything* which is, fast and loose, a theory which consistently permits to integrate the four fundamental forces which are *electromagnetic force*, *strong atomic force*, *weak atomic force* and *gravity*.

047 Regardless of the mathematical beauty of string theory, there are some downs that
048 physicists face whenever a new approach intends to formalize a general string theory.
049 An important one is that string theory, seen as a framework able to characterize an
050 unimaginable amount of possible universes; this is called *the landscape*.

051 Possible universes can be classified by it's geometry as *de Sitter*, *flat* or *anti de*
052 *Sitter*, each one with different physical structure and interactions. A *de Sitter* uni-
053 verse is one that expands at a constant acceleration, an anti *de Sitter* is an extremely
054 symmetric universe where one can return to the start point going *entlang*, and a *flat*
055 universe is an universe which is not expanding. Following Einstein's relativity our
056 universe is anti de Sitter.

057 From the landscape and geometric universe classification, *swaplant* conjectures
058 are formulated. The *swaplant* mainly indicates that string theory may not be able
059 to construct stable *de Sitter* universes.

060 Other of the interesting observed behaviour in our universe is the presence of two
061 type of quantum particles called *bosons* which "conform energy" and fermions which
062 "conform matter". For the current particle model, there are differences between bosons
063 and fermions, which are not necessarily symmetric; understanding as symmetry the
064 existence of an energy particle per matter particle and vice versa. This constrains the
065 search even more, so the configuration should let an universe which expands and also,
066 have the particles detected in our standard particle model. In other words, finding an
067 universe configuration which lets the λ CDM model emerge from it's structure.

068 In a previous from Cesar Damian [6] work there was made an attempt to check
069 the validity the conjectures in k-theory model finding metastable vacua, which con-
070 firms such conjectures using simulated annealing and ML algorithms. However, since
071 there are many possible configurations, there could be more possible configurations,
072 there are plenty of vacua to be mapped, and since simulated annealing was the only
073 one metaheuristic algorithm considered. The main proposal is a more algorithm-
074 focused research testing different methods in order to map the most possible universe
075 configurations.

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078 2 Methods

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080 3 Results

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082 4 Discussion

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084 5 Conclusions

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086 References

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